パルサー

Gunn & Ostreiker Model

GRO (EGRET ~GeV) 7つ

Cerenkov TeV

$$L_{md}=\frac{2\ddot{\vec{\mu}}^2}{3c^3}=\frac{2\mu^2\Omega^4\sin^2\chi}{3c^3}=9.6\times10^{30}\left(\frac{B}{10^{12}G}\right)^2\left(\frac{R}{10^6cm}\right)^6 P^{-4}\ erg/s$$

$$\mu=\frac{1}{2}BR^3$$

$$\frac{dE_{rot}}{dt}=I\Omega\dot{\Omega}=-\frac{4\pi^2 I\dot{P}}{P^3}=L_{md}\qquad \dot{P}=2.4\times10^{-16}\frac{B_*^2R_*^6}{PI_*}\qquad I_*=\frac{I}{10^{45}gm^2}$$

Age $\quad P(t)=\sqrt{P(0)^2+\frac{B_*^2R_*^6}{I_*}\frac{t}{6.5\times10^7yr}}\qquad t=\frac{P^2-P(0)^2}{2P\dot{P}}\approx\frac{P}{2\dot{P}}$

M ◯ $\quad \frac{dE_{rot}}{dt}=I\Omega\dot{\Omega}\propto\Omega^{n+1}$

$$\dot{\Omega}=k\Omega^n$$

$$\ddot{\Omega}=kn\Omega^{n-1}\dot{\Omega}$$

$$n=\frac{\Omega\ddot{\Omega}}{\dot{\Omega}^2}=3 \quad \text{for MDR} \qquad n_{obs}<3$$

light Cylinder $\quad r_{lc}=\frac{c}{\Omega}=4.8\times10^9P_*\ cm\qquad P_*=P/10ms$

$$B_{lc}\approx\frac{B}{r_{lc}^3}R^3\qquad L\sim4\pi r_{lc}^2\frac{B_{lc}^2}{8\pi}c\sim B^2\left(\frac{\Omega}{c}\right)^4R^6c\sim\frac{B^2R^6\Omega^4}{c^3}\sim L_{md}$$

◀ wave zone

$r\gg r_{lc}$

$$\vec{E}=\frac{1}{c^2r}\vec{n}\times\ddot{\vec{\mu}},\quad \vec{H}=\frac{1}{c^2r}\vec{n}\times(\vec{n}\times\ddot{\vec{\mu}})$$

$$E_\theta=H_\varphi=\frac{\mu\Omega^2}{c^2r}\sin\left\{\Omega\left(t-\frac{r}{c}\right)\right\}=2.2\times10^8\,B_*R_*^2P_*^{-2}\frac{\sin\left\{\Omega\left(t-\frac{r}{c}\right)\right\}}{r/R}$$

strengness)

$$\nu=\omega_H/\Omega=3.4\times10^9\frac{Z}{A}\frac{B_*R_*^2P_*^{-1}}{r/R}$$

$\nu\ll1$ ふつうの電波波

$\nu\gg1$ strong wave $\quad \frac{eE}{\Omega}\sim\frac{eB}{\Omega}\gg mc$

Strong wave 中の荷電粒子の運動

$$\frac{d\vec{P}}{dt}=Ze\left(\vec{E}+\frac{\vec{v}}{c}\times\vec{H}\right)$$

z方向に伝播する直線偏光平面波

$$E_x=H_y=B_0\sin\chi\qquad \chi=\Omega\left(t-\frac{z}{c}\right)$$

$$\begin{cases}\dot{P}_x=Ze\left(1-\frac{v_z}{c}\right)B_0\sin\chi & \nu=\frac{ZeB_0}{mc\Omega}\\ \dot{P}_y=0\\ \dot{P}_z=Ze\frac{v_x}{c}B_0\sin\chi\end{cases}$$

$$\frac{d\chi}{dt}=\Omega\left(1-\frac{v_z}{c}\right)\qquad P_x=-\nu mc(\cos\chi-\cos\chi_0)$$

$$\frac{d\gamma}{dt}=\frac{Ze}{mc^2}v_xB_0\sin\chi$$

$$\gamma=\frac{P_z^2}{mc}+1=\nu\frac{v_z}{c}+1\qquad \nu\left(1-\frac{v_z}{c}\right)=1\qquad \frac{d\chi}{dt}=\frac{\Omega}{\gamma}\qquad P_z=\frac{\nu^2}{2}mc(\cos\chi-\cos\chi_0)^2$$

$V \sim 10^8 \Rightarrow \gamma \sim V^2 \sim 10^{16}$ 10^{25} eV (proton) ?
10^{22} eV (electron)

$T = \int_0^T dt = \int_0^{2\pi} \frac{dx}{\Omega(1-\frac{v_2}{c})} = \frac{1}{\Omega}\int_0^{2\pi} \gamma dx = P\left\{1 + \frac{V^2}{4}(1+2\cos^2\chi_0)\right\}$

波の周期の V^2 倍 $\sim 10^{14}$ sec (伝播距離 ~ 1 Mpc) パルサーの寿命より長い

「phase locking」 nebula の拡がりよりはるかに長い

実際には一定の位相に lock されてしまう

spherical waves

$\chi = \chi_0 + \chi_1 \quad (\chi_1 \ll \chi_0)$ $\frac{P_z}{mc} \approx \gamma \approx \frac{V^2}{2}\chi_1^2 \sin^2\chi_0$

$z \approx \int c dt \approx \int c\, dx \frac{\gamma}{\Omega} \approx r_{sc}\frac{V^2\chi_1^3}{6}\sin^2\chi_0$

$\Big\} \gamma = \left(\frac{3}{\sqrt{2}}\frac{z}{r_{sc}} V \sin\chi_0\right)^{1/3}$

$z \sim r_{sc}$ くらい

$\gamma_{max} = 2.8\times10^5 \left(\frac{Z}{A}\right)^{1/3} B_*^{2/3} R_*^2 P_*^{-4/3}$

proton 3×10^5 GeV

electron 2×10^4 GeV

$L_{md} = 4.3\times10^{22}\gamma_{max}^3$ erg/s

問題点

① まわりは真空か？ $\omega_{pe} = 5.6\times10^4 n_e^{1/2}$ s^{-1}

② 波は安定に伝搬できるか？

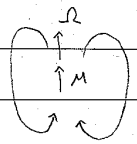
③ Crab Nebula Theory : 電子加速 $(\lesssim 10 r_{sc})$ ⇒ nonlinear syncro-Compton loss

⇒ circular polarization

obs : extended nebula

weak circular polarization

単極誘導



中性子星内部は完全導体として

$\vec{E} = -\frac{\vec{v}}{c}\times\vec{B} = \frac{1}{c}(\vec{v}\times\vec{\Omega})\times\vec{B}$

$B_r = \frac{2M\cos\theta}{r^3}$ $E_r = \frac{\Omega M}{c}\frac{\sin^2\theta}{r^2}$

$B_\theta = \frac{M\sin\theta}{r^3}$ $E_\theta = -\frac{2\Omega M}{c}\frac{\sin\theta\cos\theta}{r^2}$

$\Phi_{int} = \frac{\Omega M}{c}\left(\frac{\sin^2\theta}{r} + C_1\right)$

$\rho_e = \frac{\mathrm{div}\vec{E}}{4\pi} = -\frac{\Omega M}{2\pi c r^3}(3\cos^2\theta - 1) + \frac{2\Omega M}{3c}\delta(r)$

外部は真空とすると　重の連続性から重は4重極

$\Phi_{ext} = -\frac{\Omega M R^2}{3cr^3}(3\cos^2\theta - 1)$　　$C_1 = -\frac{2}{3R}$

$\begin{cases} E_r = -\frac{\Omega M R^2}{cr^4}(3\cos^2\theta - 1) \\ E_\theta = -\frac{\Omega M R^2}{cr^4} 2\sin\theta\cos\theta \end{cases}$　表面で E_r は不連続

表面電荷 $\sigma_e = \frac{E_{r,ext} - E_{r,int}}{4\pi} = \frac{-\Omega M \cos^2\theta}{2\pi R^2 c}$

積分すると $-\frac{2\Omega M}{3c}$ となって中心電荷を打ち消す

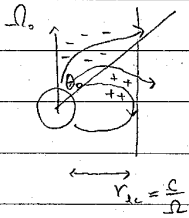
星は net charge をもたない

$\vec{E}\times\vec{B} = \frac{M^2\Omega R^2}{cr^7}\sin\theta(1+\cos^2\theta)\vec{e}_\varphi$

$\vec{v}_{drift} = \frac{c(E\times B)}{B^2} = R\Omega\sin\theta\frac{R}{r}\frac{1+\cos^2\theta}{1+3\cos^2\theta}\vec{e}_\varphi$

$\vec{E}\cdot\vec{B} = -\frac{4M^2\Omega R^2}{cr^7}\cos^3\theta$　頂点像に沿って粒子加速

$E_{//} \sim \frac{B^2R^6\Omega R^2}{BcR^7} = B\frac{R\Omega}{c}$　　$E_{//}\cdot R \sim BR\frac{R\Omega}{c} \sim 10^{16}$ c.g.s $\sim 3\cdot10^{18}$ V



$\Phi = 0$ for vacuum solution　　双極子磁力線の方程式は $\frac{r}{\sin^2\theta} = \text{const}$

$\rho_E = 0$ for plasma solution

$\sin^2\theta_0 = \frac{R\Omega}{c}$　　$\text{div}\,E = -\frac{1}{c}\text{div}((\Omega\times r)\times B)$

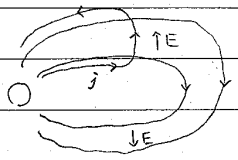
↓

* vacuum solution で $\Phi = 0$　$\cos^2\theta = \frac{1}{3}$

* plasma solution で $\rho_E = 0$

(infinite conductivity)　やはり $\cos^2\theta = \frac{1}{3}$

$\rho_E = -\frac{\vec{\Omega}\cdot\vec{B}}{2\pi c} = -\frac{M\Omega}{2\pi cr^3}(3\cos^2\theta - 1)$



$n \sim 10^{14}$ cm^{-3} for "Charge separated plasma"　×

charge neutral に近ければ n はもっと大きい

potential difference

$\Delta\Phi = \frac{M\Omega}{cR}\sin^2\theta_0 = \frac{M\Omega^2}{c^2} = 6.6\times10^{15} B_*R_*^3P_*^{-2}$ V

$L_{UI} = \pi(R\theta_0)^2 \times \frac{|\rho_E|c}{2} \times \Delta\Phi = 7.2\times10^{38} B_*^2R_*^6P_*^{-4}$ erg/s　　$P_* = P/10ms$　　MDR の Energy loss rate とほぼ等しい

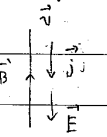
電流が頂場を横切る時に電場が仕事をしてつうずまにエネルギーを与える

Space Charge Limited Flow　真空管に電場を与える

MHD

$\frac{c}{4\pi} \mathrm{rot}\, \vec{B} = \vec{j} = \sigma \left(\vec{E} + \frac{\vec{v}}{c} \times \vec{B} \right)$

free $\sigma \to \infty$, $\frac{\vec{v}}{c} \times \vec{B} \to 0$



$\frac{dE}{dx} = 0$

$(\rho_{e,G-J} + \delta\rho_e)\, v = \left(-\frac{B\Omega}{2\pi c} + \delta\rho_e \right) v = \mathrm{const}$

$E = -\frac{d\psi}{dx}$

$\frac{dE}{dx} = -\frac{d^2\psi}{dx^2} = 4\pi\, \delta\rho_e$

$v \frac{dp}{dx} = -eE = e\frac{d\psi}{dx} = mc^2 \frac{d\gamma}{dx} \rightarrow mc^2\gamma - e\psi = \mathrm{const}$

$\frac{d\gamma}{dx} = y$ とおくと $\frac{dy}{dx} = \frac{dy}{d\gamma}\frac{d\gamma}{dx} = y\frac{dy}{d\gamma} = \frac{1}{2}\frac{dy^2}{d\gamma}$

$-4\pi e\, \delta\rho_e = mc^2 \frac{dy}{dx} = \frac{mc^2}{2}\frac{dy^2}{d\gamma}$

$x = x_0$ で $\delta\rho_e = 0$ $\quad v = v_0,\ \gamma = \gamma_0$ $\quad \delta\rho_e = \frac{B\Omega}{2\pi c}\left(1 - \frac{v_0}{v}\right)$

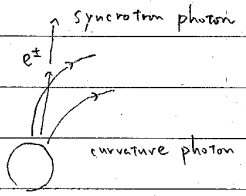
$y^2 = \frac{4eB\Omega}{mc^3}\left[\frac{v_0}{c}\sqrt{\gamma^2 - 1} - \gamma + \frac{1}{\gamma_0}\right] + y_0^2$

$x = 0$ で $y = -\frac{eE}{mc^2} = 0$

$\gamma \approx \sqrt{\frac{4eB\Omega}{mc^3}}\, x \sim \sqrt{\frac{4eB\Omega}{mc^3}}\, R\sqrt{\frac{R\Omega}{c}} \sim 3 \cdot 10^5$ for Crab

$\frac{\gamma mc^2}{e\Delta\Phi} \sim \sqrt{\frac{16mc^2}{eBR}\frac{c^2}{R^2\Omega^2}} \sim 1$ となるので

電子 $\begin{cases} \Omega \sim 10^{-2} \\ P \sim \frac{2\pi}{\Omega} \sim 500\,\mathrm{sec} \end{cases}$ $\quad$ 陽子 $\begin{cases} \Omega \sim 0.3 \\ P \sim 20\,\mathrm{sec} \end{cases}$



Single Polar Cap Model

1次電子加速 → 曲率放射 → 磁場による一光子吸収

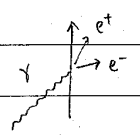
$\sim 10\,\mathrm{TeV}$

曲率放射
$\nu_c \approx \frac{c}{2\pi R_c}\gamma^3$ Hz
$-\frac{dE}{dt} \approx \frac{ce^2}{R_c^2}\gamma^4$ erg/s
$R_c = \frac{r}{3\sin\theta}\frac{(1+3\cos^2\theta)^{3/2}}{1+\cos^2\theta}$ 典型的には $\sin\theta \sim \sqrt{\frac{r_c}{R}} \ll 1$
$R_c \sim \sqrt{r_c R} \sim 10^7$ cm for Crab

$\nu_c \sim 10\,\mathrm{GeV}\left(\frac{\gamma}{10^7}\right)^3$

$\tau_{cool} \sim \frac{R_c^2 mc^2}{ce^2\gamma^3} \sim 10^{-4}\left(\frac{\gamma}{10^7}\right)^{-3}$ sec

Magnetic Pair Production



$E_\gamma = E_+ + E_-$

$\frac{E_\gamma}{c}\cos\theta = P_+\cos\theta_+ + P_-\cos\theta_-$

threshold $E_\gamma = \frac{2mc^2}{\sin\theta}$

$E_i = \sqrt{m^2c^4 + P^2c^2 + 2im^2c^4\frac{B}{B_{cr}}}$

量子力学的に干渉し

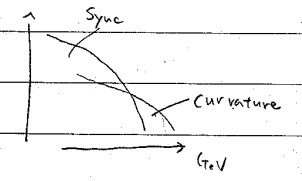
$E_\gamma = E_+ + E_-$

$\frac{E_\gamma}{c}\cos\theta = P_+ + P_-$

吸収係数 $\chi = \frac{E_\nu}{2mc^2}\frac{B}{B_{cr}}\sin\theta$ $\chi \ll 1$ に対しては

$$K(\chi) = 0.23\,\frac{2\pi\alpha_F}{\lambda_e}\frac{B}{B_c}\sin\theta\,\exp\left(\frac{-4}{3\chi}\right)$$

しかし $\frac{B}{B_{cr}} \sim 0.1$ $\sin\theta \sim 0.1$



$\frac{E_\nu}{2mc^2} \sim 10^3 \Rightarrow \chi \sim 10$

$$K(\chi) = 0.08\,\frac{2\pi\alpha_F}{\lambda_e}\frac{B}{B_c}\sin\theta\,\frac{1}{\chi}K_{1/3}^2\left(\frac{4}{3\chi}\right)F(\chi)$$

$$= A + \frac{b}{1+\exp\left[-\frac{a_1\chi - c}{B}\right]}$$

* radio, optical, X-ray の ﾋﾟｰｸﾝ

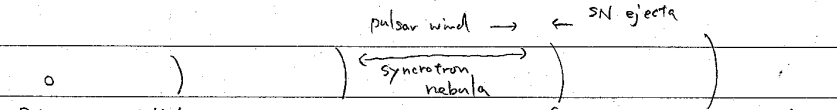
$$\frac{2\nu^2 kT_b}{c^2}\pi R_p^2 \sim L_\nu \qquad T_b \sim 10^{27}\left(\frac{1\,GHz}{\nu}\right)^3\left(\frac{L_\nu}{10^{14}\,erg/s/Hz}\right)K$$

Particle energy flux

$3\times10^{37}\,s^{-1} \times 30\,MeV \sim 10^{33}\,erg\,s^{-1}$
($e^\pm$ (2次))

1次電子 $10^{33}\,s^{-1}$, $10\,TeV \sim 10^{34}\,erg\,s^{-1}$ ほぼ γ-ray と particle に同配分

パルサー 磁気圏 · 動径依存 定常



$\sim10^6$ cm $\sim10^8$ cm ~0.1 pc ~2 pc ~5 pc

$e^\pm$ 対生成 Poynting flux

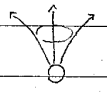
基礎方程式

$\begin{cases} \mathrm{div}(n\gamma\vec{v}) = 0 & n: \text{comving number density} \\ \gamma\vec{v}\,\mathrm{div}\left\{\left(\rho+\frac{P}{c^2}\right)\gamma\vec{v}\right\} + \left(\rho+\frac{P}{c^2}\right)\gamma(\vec{v}\,\mathrm{grad})\gamma\vec{v} + \mathrm{grad}P = \rho_e\vec{E} + \frac{\vec{j}}{c}\times\vec{B} & \text{運動方程式「一流体近似」} \\ \mathrm{div}\left\{\left(\rho+\frac{P}{c^2}\right)\gamma^2\vec{v}\right\} = \vec{j}\cdot\vec{E} & \text{エネルギー保存} \\ P \propto n^\Gamma, \quad \frac{\rho c^2}{n} = mc^2 + \frac{1}{\Gamma-1}\frac{P}{n} & \text{断熱状態方程式} \end{cases}$

$\begin{cases} \mathrm{div}\vec{E} = 4\pi\rho_e & \mathrm{div}\vec{B} = 0 \quad \vec{B} = \mathrm{rot}\vec{A} \\ \mathrm{rot}\vec{B} = \frac{4\pi}{c}\vec{j} & \vec{E} = -\mathrm{grad}\Phi \end{cases}$

$\begin{cases} \vec{E} + \frac{\vec{v}}{c}\times\vec{B} = 0 & \text{MHD 条件 (Ohm の法則)} \\ \mathrm{div}\vec{j} = 0 & \text{「局所共動系で電場が消える」} \end{cases}$

軸対称系の記述 magnetic flux



$$B_z = \frac{1}{\varpi}\frac{\partial}{\partial \varpi}(\varpi A_\varphi) = \frac{1}{2\pi\varpi}\frac{\partial f}{\partial \varpi}$$

$$B_\varpi = -\frac{\partial A_\varphi}{\partial z} = -\frac{1}{2\pi\varpi}\frac{\partial f}{\partial z}$$

$$f = 2\pi\varpi A_\varphi = \int 2\pi\varpi\, d\varpi\, B_z$$

$$j_\varphi = \frac{c}{4\pi}(\mathrm{rot}\,\vec{B})_\varphi = -\frac{c}{8\pi^2\varpi}\left[\frac{\partial^2 f}{\partial \varpi^2} - \frac{1}{\varpi}\frac{\partial f}{\partial \varpi} + \frac{\partial^2 f}{\partial z^2}\right]$$

poloidal current

$$j_z = \frac{c}{4\pi}\frac{1}{\varpi}\frac{\partial}{\partial \varpi}(\varpi B_\varphi)$$

$$j_\varpi = -\frac{c}{4\pi}\frac{\partial B_\varphi}{\partial z} \qquad J = \int 2\pi\varpi\, d\varpi\, j_z = \frac{c}{2}\varpi B_\varphi \quad \text{current flux}$$

○ electro static potential

星の表面で $\Phi = \frac{M\Omega}{3CR}(1-3\cos^2\theta) + \mathrm{const}$

一方 $f = \frac{2\pi M}{R}(1-\cos^2\theta)$ for dipole magnetic field

$\therefore \Phi = \frac{\Omega}{2\pi c} f + \mathrm{const}$ 星の内部と磁気圏との間に potential gap のないと仮定

$\vec{E} = -\frac{\Omega}{2\pi c}\,\mathrm{grad} f$

$E_\varphi = 0$

$\vec{E}\cdot\vec{B} = -(\vec{B}\cdot\mathrm{grad})\Phi = 0$ 磁力線は等ポテンシャル

charge density

$$\rho_e = \frac{\mathrm{div}\vec{E}}{4\pi} = -\frac{\Omega}{8\pi^2 c}\left(\frac{\partial^2 f}{\partial \varpi^2} + \frac{1}{\varpi}\frac{\partial f}{\partial \varpi} + \frac{\partial^2 f}{\partial z^2}\right)$$

Force-free 近似

$\rho\gamma^2 v^2 \ll \frac{B^2}{4\pi}$ の場合 $\quad \frac{\rho\gamma^2 c}{\frac{B^2}{4\pi}} \sim \frac{F_{particle}}{F_{Poynting}} \sim \frac{N\gamma^2 mc^2}{L} \sim O(10^{-3})$ $\gamma\sim 10^2$ $N\sim 10^{39}$ $L\sim 10^{38}$

○

$\rho_e\vec{E} + \frac{\vec{j}}{c}\times\vec{B} = 0$ $\quad [\mathrm{div} j = 0]$

$\vec{j}\cdot\vec{E} = 0$ $\quad \vec{j}_p \parallel \vec{B}_p$, $\vec{j}_p = \alpha\vec{B}_p$, $(\vec{B}\cdot\mathrm{grad})\alpha = 0$ $\quad \alpha = \alpha(f)$ $\quad \frac{dJ(f)}{df} = \alpha(f)$

$$\frac{\partial^2 f}{\partial \varpi^2} + \frac{1}{\varpi}\frac{\varpi^2 + \varpi_{lc}^2}{\varpi^2 - \varpi_{lc}^2}\frac{\partial f}{\partial \varpi} + \frac{\partial^2 f}{\partial z^2} - \frac{16\pi^2\varpi_{lc}^2}{\varpi^2 - \varpi_{lc}^2}\frac{J(f)}{c^2}\frac{dJ}{df} = 0$$

$\varpi_{lc} = \frac{c}{\Omega}$ $J(f)$ は星の表面の遠界分

解の例

$J = 0$ (poloidal current なし) $\quad B_\varphi = 0$ $\varpi \to \varpi_{lc}$ で $\frac{\partial f}{\partial \varpi} = 0$ $B_z = 0$ $E_\varpi = 0$ $B_\varpi = E_z$ $j_\varphi = \rho_e c$

電場が消える系の速度 $\vec{v} = c\frac{\vec{E}\times\vec{B}}{B^2} = c\vec{e}_\varphi$ light Cylinder では解は破綻している

$J \neq 0$ であっても ことが多い

$E^2 - B^2 = \frac{1}{4\pi^2}\left(\frac{1}{\varpi_{lc}^2} - \frac{1}{\varpi^2}\right)\left(\left(\frac{\partial f}{\partial \varpi}\right)^2 + \left(\frac{\partial f}{\partial z}\right)^2\right) - \frac{4J^2}{c^2\varpi^2}$ なので $\varpi > \varpi_{lc}$ では 電場が消える系が存在しない解になっている

Matter の取扱い

$\vec{E} + \frac{\vec{v}}{c} \times \vec{B} = 0 \quad E_\varphi = 0 \Rightarrow \vec{v}_p \parallel \vec{B}_p \quad \vec{v}_p = k \vec{B}_p$

$c\vec{E}_p \times \vec{B}_p + (\vec{v}_t \times \vec{B}_p + \vec{v}_p \times \vec{B}_t) \times \vec{B}_p = 0$ より $\quad v_\varphi = k B_\varphi + c \frac{(\vec{E}_p \times \vec{B}_p)_\varphi}{B_p^2} = k B_\varphi + \varpi \Omega$

一般には $\vec{j}_p \nparallel \vec{B}_p$ 電流は磁気面内で磁力線を横切る $\Rightarrow$ { flow にトルクを与え仕事をする / current closure が可能になる

磁力線方向の $\frac{1}{c}\vec{j} \times \vec{B}$ は flow の加速・減速

磁力線に垂直方向の $\rho_c \vec{E} + \frac{1}{c}\vec{j} \times \vec{B}$ は磁力線と流線の形状を決める (Grad-Shafranov eq)

Number flux

$\mathrm{div}(n\gamma\vec{v}) = 0, \ \vec{v}_p = k\vec{B}_p \quad \therefore (\vec{B} \cdot \mathrm{grad}) n\gamma k = 0$ 磁力線に沿って $n\gamma k = \mathrm{const}$

$\dot{N}(f) = \int 2\pi \varpi d\varpi \, n\gamma v_z = \int_0^f n\gamma k \, df \qquad n\gamma k = \frac{d\dot{N}}{df} \qquad (n\gamma \vec{v}_p = \frac{d\dot{N}}{df}\vec{B}_p)$

energy flux

$\mathrm{div}\{(\rho c^2 + p)\gamma^2 \vec{v}\} = \vec{j} \cdot \vec{E} = -\mathrm{div}\left\{\frac{c}{4\pi}\vec{E} \times \vec{B}\right\} \quad \therefore (\rho c^2 + p)\gamma^2 \vec{v}_p + \frac{c}{4\pi}\vec{E} \times \vec{B}_t = \frac{dL}{df}\vec{B}_p \qquad (B_\varphi < 0)$

$\vec{E} \times \vec{B}_t = -\frac{1}{c}(\vec{v} \times \vec{B}) \times \vec{B}_t = \frac{1}{c}\{\vec{v} \cdot B_t^2 - \vec{B}(\vec{v} \cdot \vec{B}_t)\} = -\frac{\varpi\Omega}{c} B_\varphi \vec{B}_p \quad \therefore (\rho c^2 + p)\gamma^2 \vec{v}_p - \frac{\varpi\Omega}{4\pi} B_\varphi \vec{B}_p = \frac{dL}{df} \cdot \vec{B}_p$

or $\quad \frac{(\rho c^2 + p)}{n} \frac{d\dot{N}}{df} - \frac{\varpi\Omega}{4\pi} B_\varphi = \frac{dL}{df}$

Angular momentum flux

$(\rho c^2 + p)\frac{\varpi \gamma^2 v_\varphi}{c^2}\vec{v}_p - \frac{\varpi}{4\pi} B_\varphi \vec{B}_p = \frac{dA}{df}\vec{B}_p$

or

$\frac{\rho c^2 + p}{nc^2} \varpi \gamma v_\varphi \frac{d\dot{N}}{df} - \frac{\varpi}{4\pi} B_\varphi = \frac{dA}{df} \qquad B_\varphi \ v_p \ v_\varphi \ n \ k$ を + で表現する

f を解かないでいえること

$v_p = k B_p = M_A v_A \ ; \ v_A = \frac{B_p}{\sqrt{4\pi(\rho + \frac{p}{c^2})\gamma^2}} \qquad k = \frac{M_A}{\sqrt{4\pi(\rho + \frac{p}{c^2})\gamma^2}} = \frac{1}{n\gamma}\frac{d\dot{N}}{df}$

$\left\{ c_s^2 = \frac{dp}{d\rho} = \frac{\Gamma p}{\rho c^2 + p} c^2 \right., \quad p = \frac{n m c_s^2}{\Gamma\left(1 - \frac{1}{\Gamma - 1}\frac{c_s^2}{c^2}\right)}, \quad \rho = nm \frac{1 - \frac{c_s^2}{\Gamma c^2}}{1 - \frac{1}{\Gamma - 1}\frac{c_s^2}{c^2}} \qquad \rho c^2 + p = \frac{nmc^2}{1 - \frac{1}{\Gamma - 1}\frac{c_s^2}{c^2}}$

$$\left\{ \begin{array}{l} k = \dfrac{\frac{1}{4\pi} M_A^2 c^2 \left(M_A^2 - 1 + \frac{\varpi^2\Omega^2}{c^2}\right)}{(M_A^2 - 1)\frac{dL}{df} + \frac{dA}{df}\Omega} \\ B_\varphi = \dfrac{4\pi}{\varpi^3} \cdot \dfrac{\frac{dA}{df} - \varpi^2\Omega\frac{dL}{df}}{M_A^2 - 1 + \frac{\varpi^2\Omega^2}{c^2}} \\ v_\varphi = \dfrac{(M_A^2 c^2 + \varpi^2\Omega^2)\frac{dA}{df} - \varpi^2\Omega\frac{dL}{df}}{\varpi\left[(M_A^2 - 1)\frac{dL}{df} + \frac{dA}{df}\Omega\right]} \end{array} \right.$$

粒は cold を仮定 $p = nm$, $P = 0$ $\quad K = M_A \frac{v_A}{\beta_p} = \frac{M_A}{\sqrt{4\pi n m \gamma^2}} = \frac{1}{n\gamma}\frac{dN}{dt}$

$n = \frac{4\pi m}{M_A^2}\left(\frac{dN}{dt}\right)^2$

$M_A \propto \frac{1}{\sqrt{n}} \propto \varpi$ (for spherical) $\quad M_A \approx \frac{\varpi}{\varpi_*}$ とすると $\quad \frac{dN}{dt}\sqrt{\frac{4\pi m}{n_*}} \approx 1$

$K \approx \frac{c^2}{4\pi \frac{dL}{dt}} \varpi^2 \left(\frac{1}{\varpi_*^2} + \frac{1}{\varpi_{lc}^2} \right) \approx \frac{1}{n_*}\left(\frac{\varpi}{\varpi_*}\right)^2 \frac{1}{\gamma}\frac{dN}{dt} \qquad \gamma mc^2 \approx \frac{\frac{dL}{dt}}{\frac{dN}{dt}} \cdot \frac{1}{1+\frac{\varpi_*^2}{\varpi_{lc}^2}}$

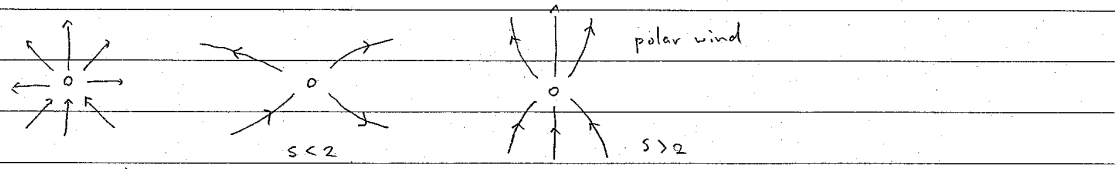
$B_\varphi \approx -\frac{4\pi}{c^2}\frac{\Omega}{\varpi}\frac{dL}{dt} \cdot \frac{1}{\left(\frac{1}{\varpi_*^2} - \frac{1}{\varpi_{lc}^2}\right)} \cdot v_\varphi = \frac{4\pi}{\varpi}$ ——

$n \propto \varpi^{-s}$ の時 (spherical でない とき) $M_A^2 \propto \varpi^s$

$s>2$ $\quad K \approx \frac{c^2}{4\pi}\frac{1}{\frac{dN}{dt}}\left(\frac{\varpi}{\varpi_*}\right)^s \approx \frac{1}{4\pi m \frac{dN}{dt}}\left(\frac{\varpi}{\varpi_*}\right)^s \frac{1}{\gamma} \qquad \gamma mc^2 \approx \frac{dL}{dt}/\frac{dN}{dt} \qquad B_\varphi \approx -\frac{4\pi}{c}\varpi\Omega\left(\frac{\varpi_*}{\varpi}\right)^s \frac{dL}{dt}$

$s<0$ $\quad k \approx$ $\qquad \gamma mc^2 \approx \left(\frac{\varpi}{\varpi_*}\right)^{s-2} \frac{c^2}{\varpi_*^2\Omega}\frac{dL/dt}{dN/dt} \to 0$

◀



sprit monopole

○

相対論的 MHD 衝撃波 martin-lees

球対称

$L = 4\pi r^2 n mc^2 \gamma^2 v_r (1+\sigma)$

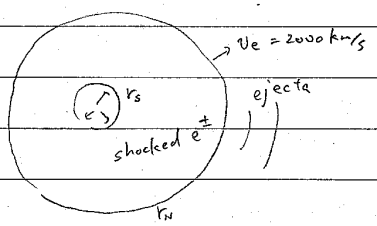
$\sigma = \frac{B_\varphi^2}{4\pi n mc^2 \gamma^2}$

$(v_\varphi \approx 0 \ , B_r \approx 0 \ , p \approx 0)$

上流で定義

relativistic $v_1 \approx c$ $\gamma_1 \gg 1$

cold $P_1 \approx 0$



$\frac{L}{4\pi c r_s^2} \sim \frac{Lt}{\frac{4\pi}{3} r_N^3}$ $\qquad r_N = v_{ej} t$

$r_s \sim v_{ej} t \sqrt{\frac{v_{ej}}{c}}$

0.1 pc $\quad 10^3$ yr

Rankine-Hugoniot relations

$n_1 \gamma_1 v_1 = n_2 \gamma_2 v_2$ (number flux)

$E_\theta = \frac{v_1}{c} B_1 = \frac{v_2}{c} B_2$ (electronic field)

$(\rho_1 c^2 + P_1)\gamma_1^2 v_1 + \frac{E_\theta B_1}{4\pi}$

$\frac{B_2}{B_1} = \frac{c}{v_2}$ (下流)

$\frac{n_2}{n_1} = \frac{\gamma_1 c}{\gamma_2 v_2}$

$\beta_2 = \frac{1+2\sigma+\sqrt{1+16\sigma+16\sigma^2}}{6(1+\sigma)}$

$P_2 = \frac{n_1 mc^2 \gamma_1^2}{4\gamma_2^2\beta_2}\left(4\sigma\left(1-\frac{1}{\beta_2}\right)\right)$

$kT_2 = \frac{P_2}{n_2} = mc^2 \frac{\gamma_1}{4\gamma_2}\left[1+\sigma\left(1-\frac{1}{\beta_2}\right)\right]$

$\sigma \approx 3\times 10^{-3}$

シンクロ frequency $\frac{eB}{mc\delta}$

$\sigma \gg 1$ Pynting flux dominated → flow は relativistic

$\sigma \ll 1$ Particle flux dominated → 〃 mildly relativistic

下流 $\beta_\infty = \frac{\beta_2}{1+\Delta_2} = \frac{\sigma}{1+\sigma}$ $n \propto \frac{1}{r^2}$ $P \propto r^{-\frac{8}{3}}$ $B_\varphi \propto \frac{1}{r}$

遠方では再び Pynting flux dominated

$\Delta_2 = \frac{16\pi\gamma_2^2 P_2}{B_{\varphi 2}^2} = \frac{1+\sigma}{\sigma}\beta_2 - 1$

$L \sim 3\times10^{38}$ erg/s スピンダウン $\sigma = 3\cdot10^{-3}$ とすると

$B_{\varphi 2} \sim 10^{-4}$ G $n\gamma_1^2 \sim 0.03$

$B_{\varphi 1} \sim 3\times10^{-5}$ G $\sim E_0 \sim \frac{\Delta P}{\Delta r} \sim 10^{-5}$ cgs $L \sim 10^{39}$ erg/s } となる

$L \sim 3\cdot10^{32}$ erg/s

$\dot{N} \sim 3\cdot10^{37}$ /s

$\gamma mc^2 \sim L/\dot{N} \sim 10$ erg $[\gamma_1 \sim 10^7]$

$N \sim \dot{N}\tau \sim 3\cdot10^{37} \cdot 3\cdot10^{9} \sim 10^{48}$ コ

13:15

§3 フェルミ加速

(1) Original な Fermi 加速
(2) 衝撃波による加速
(3) 斜め 〃

衝撃波のいろいろある SNR, Radio Gal

正面衝突 $2p\frac{V}{v}$ ← cloud / particle だけ運動量増加 確率は $\frac{v+V}{L}$
追突 〃 減少 〃 $\frac{v-V}{L}$

$$\frac{dP}{dt} = 2p\frac{V}{v}\left(\frac{v+V}{L} - \frac{v-V}{L}\right) = 4p\frac{V}{L}\left(\frac{V}{v}\right)^2 \qquad p = p_0 e^{t/\tau_A}$$

粒子の寿命の分布 { 原子核衝突 / 磁場からの逃散 } τ_L (Loss time) $f(t) = \frac{1}{\tau_L}e^{-t/\tau_L}$

粒子の運動量(エネルギー)分布

$$n(p) = f(t)\frac{dt}{dp} = \frac{1}{\tau_L}e^{-t/\tau_L}\frac{\tau_A}{p} = \frac{\tau_A}{\tau_L}\frac{1}{p_0}\left(\frac{p}{p_0}\right)^{-\frac{\tau_A}{\tau_L}-1}$$

問題点 ① $\tau_A \sim \tau_L \sim 10^7$ yrs

$L \sim 30$ pc とすると $\frac{V}{c} \sim 2\cdot10^{-3}$

$V \sim 500$ km/s 必要

$V \sim 10$ km/s なら $\tau_A \sim 10^{18}$ sec であり

加速されない

② SNR など V の大きい場所は τ_L/τ_A は一定というのもへん

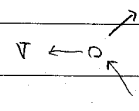
Fokker - Planck 方程式

$$f(\vec{p}, t+\Delta t) = \int d\Delta\vec{p}\, \psi(\vec{p}-\Delta\vec{p}, \Delta\vec{p})\, f(\vec{p}-\Delta\vec{p}, t) = \int d\Delta\vec{p} \left[\psi f - \Delta p \frac{\partial}{\partial \Delta p} \psi f + \frac{\Delta p \Delta p}{2} \frac{\partial^2}{\partial \Delta p \partial \Delta p} \psi f \right]$$

$$\frac{\partial f}{\partial t} = -\frac{\partial}{\partial p} \left\{ \left\langle \frac{\Delta p}{\Delta t} \right\rangle f \right\} + \cdots$$

$\psi(\vec{p}, -\Delta\vec{p}) = \psi(\vec{p}-\Delta\vec{p}, \Delta\vec{p})$ detailed balance $\quad \frac{\partial}{\partial p} \left\langle \frac{\Delta p}{\Delta t} \right\rangle - \frac{1}{2} \frac{\partial^2}{\partial p \partial p} \left\langle \frac{\Delta p \Delta p}{\Delta t} \right\rangle = 0$

(注) $\frac{\partial f}{\partial t} = \frac{\partial}{\partial p} \left\{ D_{pp} \frac{\partial f}{\partial p} \right\}$ $\quad D_{pp} = \frac{1}{2} \left\langle \frac{\Delta p \Delta p}{\Delta t} \right\rangle$



1次で $\Delta p \approx \frac{V_2}{pc^2}(p_x' - p_x)$ $\quad D_{pp} = \frac{1}{2} \left\langle \frac{\Delta p \Delta p}{\Delta t} \right\rangle = \frac{1}{2} \frac{V^2}{v^2} \langle p_x'^2 + p_x^2 \rangle \frac{\nu}{L} = \frac{1}{3} \frac{p^2 V}{\nu L}$

Leaky Box Model

$\frac{\partial f}{\partial t} = \frac{1}{p^2} \frac{\partial}{\partial p} \left(p^2 D_{pp} \frac{\partial f}{\partial p} \right) - \frac{f}{\tau_L}$ steady state solution $\tau_L = \text{const}$ $fp^2 \propto p^{-n}$ と仮定すると

power law solution

$n = \frac{3}{2} \sqrt{1 + \frac{\tau_A}{\tau_L}} - \frac{1}{2}$ が存在

τ_L が p 依存性をそれば べき解はない

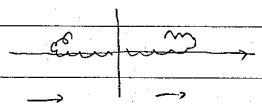
$\tau_L \to \infty$ では $f \propto p^{-3}$ $\quad fp^2 \propto p^{-1}$

$\tau_L \to 0$ では exp cutoff

$\tau_A = \frac{4cL}{3V^2}$ $\quad (v = c$ とした)

2次の Fermi 加速

(2)



仮定 plane-parallel, steady-state

non-relativistic shock

$\vec{B}$ は shock normal 方向 (parallel shock)

非熱的粒子は Alfven 波によって弾性散乱される

衝撃波 given

up stream shockfront down stream

* individual particle approach

$V \ll c$, $E = cp$

上流→下流 $\quad \bar{E}_k^A = E_k^u \left(1 + \frac{V_1}{c} \cos\theta_k^A\right) = E_k^d \left(1 + \frac{V_2}{c} \cos\theta_k^A\right)$

k回目

下流→上流 $\quad \bar{E}_k^B = E_k^d \left(1 + \frac{V_2}{c} \cos\theta_k^B\right) = E_{k+1}^u \left(1 + \frac{V_1}{c} \cos\theta_k^B\right)$

$$\bar{E}_{k+1}^A = E_{k+1}^u \left(1 + \frac{V_1}{c} \cos\theta_{k+1}^A\right) = E_k^d \left(1 + \frac{V_2}{c} \cos\theta_k^B\right) \frac{1 + \frac{V_1}{c} \cos\theta_{k+1}^A}{1 + \frac{V_1}{c} \cos\theta_k^B} = \bar{E}_k^A \frac{1 + \frac{V_2}{c} \cos\theta_k^B}{1 + \frac{V_2}{c} \cos\theta_k^A} \cdot \frac{1 + \frac{V_1}{c} \cos\theta_{k+1}^A}{1 + \frac{V_1}{c} \cos\theta_k^B}$$

$$\ln \frac{\bar{E}_\ell^A}{\bar{E}_0^A} = \sum \ln \frac{\bar{E}_k^A}{\bar{E}_{k-1}^A} = \sum \ln \frac{1 + \frac{V_1}{c} \cos\theta_k^A}{1 + \frac{V_2}{c} \cos\theta_{k-1}^A} + \ln \frac{1 + \frac{V_2}{c} \cos\theta_{k-1}^B}{1 + \frac{V_1}{c} \cos\theta_{k-1}^B} \approx \ell \left[\frac{V_1 - V_2}{c} \langle \cos\theta^A \rangle - \frac{V_1 - V_2}{c} \langle \cos\theta^B \rangle \right] = \frac{4}{3} \ell \frac{V_1 - V_2}{c}$$

$$\langle \cos\theta^A \rangle = \frac{\int \cos^2\theta \, d\cos\theta}{\int d\cos\theta}$$

横切る回数の分布　数密度 n　上流からの flux $\frac{nc}{4}$　下流での net flux nV_2

上流→下流へ横切った粒子のうち $\frac{4V_2}{c}$ はそどうらい　$1-\frac{4V_2}{c}$ は再び上流へそどう

l 回だけ往復する確率 P_l

$$P_l = \left(1-\frac{4V_2}{c}\right)^l \cdot \frac{4V_2}{c} \approx e^{-\frac{4V_2}{c}l}\frac{4V_2}{c} = \frac{4V_2}{c}\exp\left(-\frac{4V_2}{c}\frac{3}{4}\frac{c}{V_1-V_2}\ln\frac{E}{E_0}\right) = \frac{4V_2}{c}\left(\frac{E}{E_0}\right)^{-\frac{3V_2}{V_1-V_2}}$$

$$N(E)dE = P_l\,dl = P_l\frac{dl}{dE}dE \qquad N(E)\propto\left(\frac{E}{E_0}\right)^{-\frac{3V_2}{V_1-V_2}-1} = \left(\frac{E}{E_0}\right)^{-\frac{r+2}{r-1}} \qquad r=\frac{V_1}{V_2}$$

universal

$r=4 \Rightarrow N(E)\propto E^{-2}$

＊ diffusion - convection equation approach　拡散対流方程式

○ $\frac{\partial f}{\partial t}+V\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(K\frac{\partial f}{\partial x}\right)+\frac{1}{3}\frac{\partial V}{\partial x}p\frac{\partial f}{\partial p}+Q$　　f：位相空間密度

対流　拡散　加速減速　源，逃散　　p：流体系での運動量

保存形 → $\frac{\partial f}{\partial t}+\frac{\partial}{\partial x}\left\{Vf-K\frac{\partial f}{\partial x}\right\}+\frac{1}{p^2}\frac{\partial}{\partial p}\left\{p^3\left(-\frac{1}{3}\frac{\partial V}{\partial x}\right)f\right\}=Q$

$n=4\pi\int_0^\infty fp^2dp$　$\frac{\partial n}{\partial t}+\frac{\partial}{\partial x}\left\{Vn-K\frac{\partial n}{\partial x}\right\}-\frac{1}{3}\frac{\partial V}{\partial x}\left[p^3f\right]_0^\infty = \int_0^\infty Q4\pi p^2dp$　$\frac{\partial V}{\partial x}=(V_1-V_2)\delta(x)$

~~$\frac{\partial n}{\partial t}+\frac{\partial}{\partial x}\left\{Vn-K\frac{\partial n}{\partial x}\right\}=4\pi p_0^2Q_0\delta(x)$~~　$Q=Q_0\delta(x)\delta(p-p_0)$

定常解 $\frac{\partial}{\partial x}\left[V_Af-K\frac{\partial f}{\partial x}\right]=0$

上流　$f_1 = g_1(p)\exp\left(\int_0^x\frac{V_1}{K(x,p)}dx\right)+g_3(p)$　$(x<0)$　$g_3(p)$ は pre-existing component

下流　$f=g_2(p)$ のみ　$(x>0)$　無限遠で発散しない

× f の連続　$g_1(p)+g_3(p)=g_2(p)$

× flux の連続　$K\frac{\partial f_1}{\partial x}+V_1\frac{p}{3}\frac{\partial f_1}{\partial p}=V_2\frac{p}{3}\frac{\partial f_2}{\partial p}$　$(x=0)$　$(p\neq p_0)$　Compton - Getting flux

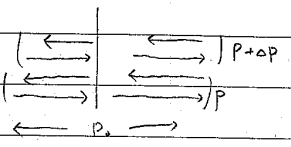
○

$\frac{dg_1}{dp} = -\frac{3V_1}{V_1-V_2}\frac{g_1}{p}-\frac{dg_3}{dp}$　解 $g_1 = p^{-\frac{3V_1}{V_1-V_2}}\int^p p^{\frac{3V_1}{V_1-V_2}}\frac{dg_3}{dp}dp + \text{const}\times p^{-\frac{3V_1}{V_1-V_2}}$

$g_2=g_1+g_3$

$= \frac{3r}{r-1}p^{-\frac{3r}{r-1}}\int^p p^{\frac{2r+1}{r-1}}g_3(p)dp$ ⇐ incoming particle

$+\text{const}\times p^{-\frac{3r}{r-1}}$ ⇐ newly injected part



下流での flux　$-\frac{V_2}{3}p\frac{\partial f}{\partial p}$　積分すると　$-\frac{V_2}{3}\int 4\pi p^3\frac{\partial f}{\partial p}dp = V_2n$

上

$-K_1\frac{\partial f}{\partial x}-\frac{V_1}{3}p\frac{\partial f}{\partial p}$　0

加速時間

上流滞在時間 $N = \int_{-\infty}^{\infty} n e^{V_1 x/K_1} dx = \frac{K_1 n}{V_1}$ $\quad F = \frac{nc}{4}$ $\quad t_1 = \frac{N}{F} = \frac{4K_1}{V_1 c}$

下流 〃 $V_2 \frac{\partial n}{\partial x} = \frac{\partial}{\partial x}\left(K_2 \frac{\partial n}{\partial x}\right) + Q\,\delta(x - x_0)$ を考える

境界条件 $n(0) = 0$, $n(\infty)$ 有限

解 $n = \begin{cases} \frac{Q}{V_2}\left[e^{\frac{V_2}{K_1}x} - 1\right] e^{-\frac{V_2 x_0}{K_1}} & (0 \le x \le x_0) \\ \frac{Q}{V_2}\left[1 - e^{-\frac{V_2 x}{K_1}}\right] & (x_0 < x < \infty) \end{cases}$

往復時間 K_1, K_2 拡散係数

$\Delta t = t_1 + t_2 = \frac{4}{c}\left(\frac{K_1}{V_1} + \frac{K_2}{V_2}\right)$

$\Delta p = \frac{4}{3}\frac{V_1 - V_2}{c} p$ $\qquad t_{acc} = \frac{p\Delta t}{\Delta p} = \frac{3}{V_1 - V_2}\left(\frac{K_1}{V_1} + \frac{K_2}{V_2}\right)$ 「加速率は V の2次である」

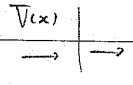
位相空間密度 $f(\vec{x}, \vec{p}, t)$

数密度 $n(\vec{x}, p, t) = p^2 \int d\hat{p}\, f(\vec{x}, \vec{p}, t)$

流束密度 $\vec{S}(\vec{x}, p, t) = \frac{p^3 c^2}{E} \int d\hat{p}\, \hat{p}\, f(\vec{x}, \vec{p}, t)$

保存則 $\frac{\partial n}{\partial t} + \mathrm{div}\,\vec{S} + \frac{\partial F}{\partial p} = 0$

F: 運動量空間での流束 $F = \left\langle \frac{dp}{dt} \right\rangle n$



$\hat{p} = \frac{\vec{p}}{p}$

shock系

流体系

流体系

$\vec{V}(\vec{x})$ $\quad \vec{p}_* = \vec{p} - \frac{E}{c^2}\vec{V}$ $\quad \left[\frac{V}{c}\text{ の1次まで}\right]$

$f(\vec{p}) = f_*(\vec{p}_*) \approx f_*(\vec{p}) - \frac{E}{c^2}\vec{V}\frac{\partial f_*(\vec{p})}{\partial \vec{p}}$

体では等方に近い

$f_*(\vec{p}_*) = \frac{1}{4\pi p_*^2}\left[n_*(p_*) + \frac{3E_*}{p_* c^2}\hat{p}\cdot\vec{S}_*(p_*)\right]$

Fick's law (diffusion tensor)

$\vec{S}_*(p_*) = -\overleftrightarrow{K}\,\mathrm{grad}\, n_*(p_*)$

慣性系

$f(\vec{p}) = \frac{1}{4\pi p^2}\left[n_*(p) - \frac{3E}{pc^2}\hat{p}\,(\overleftrightarrow{K}\,\mathrm{grad})\, n_*(p)\right] - \frac{E}{4\pi c^2}\vec{V}\cdot\hat{p}\frac{\partial}{\partial p}\left[\frac{n_*(p)}{p^2} - \frac{3E\hat{p}\,(\overleftrightarrow{K}\,\mathrm{grad})n_*(p)}{p^2 c^2}\right]$

↘ neglect

$n(p) = n_*(p)$

$\vec{S}(p) = -(\overleftrightarrow{K}\cdot\mathrm{grad})\, n(p) - \frac{p^3}{3}\vec{V}\frac{\partial}{\partial p}\left(\frac{n}{p^2}\right)$

運動量変化

圧力 $\underline{P} = \int d\hat{p}\cdot p^2 \cdot p\hat{p}\,\frac{pc^2\hat{p}}{E} f(p) = \frac{1}{3}\frac{p^2 c^2}{E} n(p)$

流体のなす仕事率

$W = \vec{V}\cdot\mathrm{grad}\,\underline{P} = \frac{1}{3}\frac{p^2 c^2}{E}(\vec{V}\cdot\mathrm{grad})\, n(p)$ $\quad F = \left\langle \frac{dp}{dt} \right\rangle n = \frac{E}{c^2 p}\left\langle \frac{dE}{dt} \right\rangle n =$

結局 $\frac{\partial n}{\partial t} + \mathrm{div}\left[-(\overleftrightarrow{K}\,\mathrm{grad})\, n - \frac{p^3}{3}\vec{V}\frac{\partial}{\partial p}\left(\frac{n}{p^2}\right)\right] + \frac{1}{3}\frac{\partial}{\partial p}\left[(\vec{V}\cdot\mathrm{grad})\, p n\right] = 0$

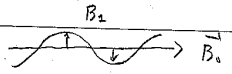
$\frac{\partial n}{\partial t} + (\vec{V}\cdot\mathrm{grad})\, n - \frac{\mathrm{div}\,\vec{V}}{3}$ …

拡散係数 pitch angle scattering

$K_{\parallel} = \frac{1}{3} v \lambda$ $\quad \Omega_L =$

$\nu = \frac{\pi}{4} \left(\frac{k \varepsilon_k}{B^2/8\pi} \right) \Omega_L$

$K_{\parallel} = \frac{r_L c}{3} \xi \qquad \xi = \frac{4}{\pi} \frac{B^2/8\pi}{k \varepsilon_k} \gtrsim 1$ Bohm limit



Alfven波の固有系 ($\vec{E_1} = 0$) 円偏光, 波数 k

0次 $p_{\parallel} = p\mu \quad p_{\perp} = p\sqrt{1-\mu^2}$

1次 $p \frac{d\mu}{dt} = \frac{q}{c} [\vec{v_{\perp}} \times \vec{B_1}]_{\parallel}$

$\vec{B_1} = B_1 [\cos(kz)\vec{e_x} + \sin(kz)\vec{e_y}]$

$\vec{v_{\perp}} = \frac{pc^2\sqrt{1-\mu^2}}{E} [\cos(\Omega t + \psi)\vec{e_x} + \sin(\Omega t + \psi)\vec{e_y}]$

$z = v\mu t$ $\quad \Omega = \frac{qB_0}{mc\gamma} = \frac{qB_0 c}{E}$

$\frac{d\mu}{dt} = \frac{cqB_1}{E}\sqrt{1-\mu^2}\sin\{(kv\mu - \Omega)t - \psi\}$

$\langle \Delta\mu \rangle = \frac{1}{2\pi}\int_0^{2\pi} d\psi \int_0^t \frac{d\mu}{dt} dt = 0$

$\langle \Delta\mu^2 \rangle = \frac{1}{2\pi}\int_0^{2\pi} d\psi \left[\int_0^t \frac{d\mu}{dt} dt\right]^2 = \frac{c^2q^2B_1^2(1-\mu^2)}{2E^2}\int_0^t dt_1 \int_0^t dt_2 \cos\{(kv\mu - \Omega)(t_1 - t_2)\}$

$t \to \infty$

$\langle \Delta\mu^2 \rangle = \frac{c^2q^2B_1^2(1-\mu^2)}{2E^2} 2\pi\delta(kv\mu - \Omega)t$

$\nu = \left\langle \frac{\Delta\theta^2}{\Delta t} \right\rangle = \frac{\pi q^2 B_1^2}{Ep\mu}\delta\left(k - \frac{\Omega}{v\mu}\right)$ $\quad 2 \times \frac{B_1^2}{4\pi}\delta\left(k - \frac{\Omega}{v\mu}\right) = \varepsilon_k$ $\quad \nu = \frac{\pi}{4}\Omega \frac{\varepsilon_k \cdot k}{\frac{B_0^2}{8\pi}}\Big|_{k=\frac{\Omega}{v\mu}}$

$\frac{\partial}{\partial p} D_{pp} \frac{\partial f}{\partial p} = \frac{\partial}{\partial \mu}\left[\frac{1-\mu^2}{2}\nu \frac{\partial f}{\partial \mu}\right]$ $\quad K_{\parallel} = \left\langle \frac{1-\mu^2}{2}\frac{v^2}{\nu} \right\rangle_{\mu} \sim \frac{1}{3}\frac{v^2}{\nu} \sim \frac{v^2}{3}\frac{4}{\pi\Omega}\frac{B_0^2/8\pi}{\varepsilon_k \cdot k} = \frac{v r_L}{3}\xi$

$\xi = 1$ Bohm limit $\qquad \xi = \frac{4}{\pi}\frac{B_1^2/8\pi}{\varepsilon_k \cdot k} \gtrsim 1$

到達エネルギー

$\begin{cases} t_{acc} \propto \frac{K}{v^2} \propto \frac{\gamma}{v^2} \\ t_{ev} \propto \frac{R}{v} \end{cases}$ $\quad t_{acc} = t_{ev} \Rightarrow \gamma_{max}$ $\quad E_{max} = \frac{3}{20}\frac{1}{\xi}\frac{\bar{v}_1}{c} ZeBR$

(t_{ev}: evolution (life time))

$= 1.35 \times 10^{14} \times Z \frac{1}{\xi}\frac{\bar{v}_1}{c}\left(\frac{B}{1\mu G}\right)\left(\frac{R}{1pc}\right) e\bar{V}$

このエネルギーは十分大きいか?

$E < E_{max}$ では $t_{acc} < t_{ev} \Rightarrow K < R\bar{v}$

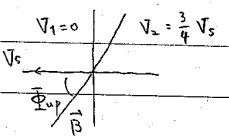
$t_{acc} < t_{ev} < t_{diff}$ $\quad t_{diff} \sim \frac{R^2}{K} \sim \frac{R}{c}\cdot\frac{R}{\lambda} > \frac{R}{v} \sim t_{ev}$

E^u up stream

E^d down stream

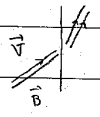
4. 斜め衝撃波

Inertia Frame

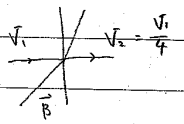


time - dependent

de-Hoffman Teller Frame



Shock Frame



衝撃波面と磁力線の交点が動く

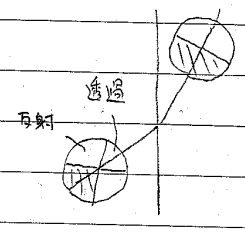
Eが現れる

$\vec{B} \parallel \vec{V}$

$\vec{E}=0$

$\frac{V_s}{c} < \cos\theta_{up}$ の時にこのFrameをとることが可能 (subluminal shock)

磁場のdynamicalな影響は無視



Lorentz変換

$V_t = -V_s$ $t_{an}\theta_1^s$ =

$|V_t| > c \Leftrightarrow \frac{V_s}{c} \cos\theta$: superluminal shock

subluminal shock

粒子の反射と透過

断熱近似 $\begin{cases} p^2 \text{ 保存} \\ \frac{B^2}{B} = \frac{p^2(1-\mu^2)}{B} \text{ 保存} \end{cases}$

$b \equiv \frac{B_2}{B_1} = \sqrt{r^2-(r^2-1)\beta_s^2\left(\cos^2\theta_{up} - \frac{V_s^2}{c^2}\right)} = \sqrt{1+(r^2-1)\beta_s^2\sin^2\theta_{up}}$

上流→下流

$0 \le \mu \le \mu_0$ の粒子は反射される (反射後 $-\mu_0 < \mu < 0$)

$\mu_0 \le \mu$ の粒子は透過する (透過後 $0 \le \mu \le 1$)

$\frac{1-\mu_0^2}{B_1} = \frac{1}{B_2}$ $\mu_0 = \sqrt{1-\frac{1}{b}}$

下流→上流

$-1 \le \mu \le 0$ の粒子は透過する (透過後 $-1 \le \mu \le -\mu_0$)

衝撃波面の向きのflux

$\cos\theta_1 \frac{\mu d\mu}{(1-\mu V_1)^2}$ に比例

透過率 $P_* = 1-\mu_0^2$ 透過粒子 $\langle \mu_B \rangle = \frac{2}{3}\frac{1-\mu_0^3}{1-\mu_0^2}$

反射率 $P_r = \mu_0^2$ 反射粒子 $\langle \mu_B \rangle = \frac{2}{3}\mu_0$ $(\langle \mu_A \rangle = -\frac{2}{3}\mu_0)$

上流から衝撃波面にはいる粒子についての平均のエネルギー利得

反射

平均のエネルギー利得 $\Delta E = \frac{4}{3}\frac{V_1 - V_2(1-\mu_0^2)}{c}E$

逃げる確率 $(1-\mu_0^2)\frac{4V_2}{c}$

スペクトル指数 $\alpha = \frac{3V_2(1-\mu_0^2)}{V_1 - V_2(1-\mu_0^2)} + 1 = \cdots = \frac{r+2}{r-1}$ で parallel shock と同じ

加速時間

$$\begin{cases} t_1 = \dfrac{4K_1}{V_1 c} \\ t_2 = \dfrac{4K_2}{V_2 c} \end{cases} \qquad \Delta t = P_r t_1 + P_t (t_1 + t_2) = \frac{4}{c}\left(\frac{K_1}{V_1} + \frac{K_2}{V_2}(1-\mu_0^2)\right)$$

$$t_{acc} = \frac{E_0 t}{\Delta E} = \frac{3}{V_1 - V_2(1-\mu_0^2)}\left(\frac{K_1}{V_1} + \frac{K_2(1-\mu_0^2)}{V_2}\right) = \frac{3\cos^2\bar{\theta}_{up}}{V_s^2}\frac{r}{1-r}\left(K_1 + rK_2(1-\mu_0^2)^2\right)$$

* $\cos^2\bar{\theta}_{up}$ だけ加速時間が短縮される

* $(1-\mu_0^2)^2 = \dfrac{1}{1+(r^2-1)P_s^2\sin^2\bar{\theta}_{up}} \sim \dfrac{1}{r^2}$ の寄与も大きい

$K_2 \sim \frac{1}{B}$ なら $K_2 = K_1(1-\mu_0^2)$ も有効

A4 2枚 7/19 締切 プレリプ 名前 じむ かをなべ土